DECK 05 · WEEK 3 · MORNING 1

Macro Policy, Geopolitics, Trade & FX

Six CFA modules on the macro forces that drive every cross-border price

CFA Level I · Vol 2 Learning Modules 3–8

Roadmap — Macro Policy, Geopolitics, Trade & FX

6 sections · ~2 hours · Built bottom-up from CFA readings

01	**Fiscal policy toolkit + multipliers** Roles · automatic stabilizers · multiplier 1/[1−c(1−t)] · debt sustainability · implementation lags
02	**Monetary policy transmission** OMO · policy rate · reserve req · 4 channels · inflation targeting · Taylor rule
03	**Geopolitics for investors** State vs non-state actors · cooperation spectrum · event / exogenous / thematic risk
04	**Trade theory + BoP basics** Comparative advantage · tariffs & deadweight loss · trading blocs · BoP identity
05	**FX market structure & quoting** USD 7.5tn daily · USD dominance · price/base convention · bid-offer · 8 regimes
06	**FX arithmetic: cross-rates, IRP, PPP** Cross-rate chain · triangular arb · covered IRP · PPP · forward premium/discount

Learning objectives

By the end of this deck you should be able to:

1. Compute the Keynesian fiscal multiplier for a given MPC and tax rate and explain why it collapses toward one as slack in the economy disappears.

2. Describe the three monetary-policy tools and the four transmission channels, and apply the Taylor rule to infer an appropriate policy rate.

3. Classify a country on the political-cooperation spectrum and distinguish event, exogenous, and thematic geopolitical risks for portfolio monitoring.

4. Apply the principle of comparative advantage and calculate the consumer-surplus, producer-surplus, revenue, and deadweight-loss components of a tariff.

5. Read an FX quote using the price-currency / base-currency convention, derive a cross-rate through the USD, and spot a triangular-arbitrage opportunity.

6. Write down the covered interest-rate-parity condition, compute a forward rate from spot and two interest rates, and explain why PPP is a long-run anchor only.

SECTION 01

Fiscal policy toolkit

Fiscal policy uses government spending and taxation to steer aggregate demand, redistribute income, and allocate resources across sectors

Budget deficit widens in downturns (expansionary), narrows in booms (contractionary). Automatic stabilizers do this without any policy-maker decision.

ROLE 1 · DEMAND MANAGEMENT

Boost aggregate demand through higher G or lower T when output is below potential. Contractionary policy cools an overheating economy.

ROLE 2 · REDISTRIBUTION

Progressive income taxes and targeted transfers compress the income distribution; regressive VAT widens it. Political choice, not pure macro.

ROLE 3 · ALLOCATION

Subsidies, carbon taxes, infrastructure spending steer resources toward favored sectors — externality correction or industrial policy.

AUTOMATIC STABILIZERS · KEYNESIAN vs MONETARIST DEBATE

Progressive taxes and unemployment benefits cushion shocks without any policy decision — the deficit widens automatically when GDP falls. Keynesians argue fiscal stimulus is powerful when there is slack; monetarists argue it is temporary and monetary policy is the superior stabilization tool. The 2008–09 GFC response and post-pandemic stimulus reignited this long-running debate.

Source: CFA Institute, Curriculum Vol 2, Learning Module 3, §1 "Roles and Objectives of Fiscal Policy", pp. 82–90.

The fiscal multiplier scales every dollar of government spending into a larger change in aggregate output — but only when there is slack in the economy

At full employment the multiplier shrinks toward 1 as extra G crowds out private investment and raises prices rather than real output.

$$\text{Multiplier} = 1 / [\,1 - c \times (1 - t)\,]$$

where: c = marginal propensity to consume (MPC) out of disposable income; t = marginal tax rate on income.

UK FISCAL STIMULUS — KEYNESIAN MULTIPLIER CALCULATION

The UK Treasury announces a GBP 10 billion infrastructure package. With MPC c = 0.9 and tax rate t = 0.2, what is the total impact on aggregate output?

1. Disposable-income leakage: each GBP 1 of income keeps (1 − t) = 0.80 after tax.
2. Consumption out of disposable: c × (1 − t) = 0.9 × 0.8 = 0.72.
3. Multiplier = 1 / (1 − 0.72) = 1 / 0.28 = 3.57.
4. Total ΔY = 3.57 × GBP 10 bn = GBP 35.7 bn of additional aggregate output.

ΔY ≈ GBP 35.7 bn — provided there is idle capacity; if full employment, the real-output boost collapses and extra G leaks into inflation.

Source: CFA Institute, Curriculum Vol 2, Learning Module 3, §2 "The Fiscal Multiplier", pp. 102–106.

Fiscal policy operates through four tools with very different lags and effectiveness — spending, transfers, direct taxes, and indirect taxes

Spending has the largest multiplier (direct demand) but the longest implementation lag; tax cuts are fastest to implement but leakiest (savings).

Tool	Typical multiplier	Lag to implement	Main drawback
Direct government spending (G)	1.0 – 2.5	12 – 24 months	Slow — planning, procurement, political
Transfers (unemployment, pensions)	0.8 – 1.5	Weeks	Leaks into saving by high-income recipients
Direct taxes (income / corporate)	0.6 – 1.2	3 – 9 months	Weak for cash-constrained consumers
Indirect taxes (VAT, excise)	0.4 – 0.9	Weeks	Regressive; distorts relative prices

THREE IMPLEMENTATION LAGS · RECOGNITION → ACTION → IMPACT

Recognition lag: time to realize a shock has occurred. Action lag: time for legislatures to pass policy. Impact lag: time for private agents to respond. Each is 3–12 months — by the time discretionary stimulus arrives, the shock it was meant to address may have passed.

Source: CFA Institute, Curriculum Vol 2, Learning Module 3, §2 "Fiscal Policy Tools", pp. 94–102.

Debt sustainability depends on whether real GDP growth exceeds the real interest rate on outstanding government debt

If real g > real r, the debt-to-GDP ratio stabilizes or falls even with primary deficits. Japan has sustained 237% debt-to-GDP precisely because r is near zero.

$$\Delta\,(\text{Debt} / \text{GDP}) \approx (r - g) \times (\text{Debt} / \text{GDP}) - \text{Primary surplus} / \text{GDP}$$

where: r = real interest rate on debt; g = real GDP growth; primary surplus excludes interest payments.

Country	Debt/GDP 1995	Debt/GDP 2015	Current-year Δ	Stance
Japan	94.7%	237.4%	+15 p.p.	Unsustainable on face; BoJ holds ~50% of JGBs
United States	83.2%	125.3%	+42 p.p.	Dollar privilege insulates financing
United Kingdom	51.4%	111.7%	+60 p.p.	Post-GFC + Covid stimulus
Germany	54.1%	78.9%	+25 p.p.	Anchored by debt-brake rule

SECTION 02

Monetary policy transmission

Central banks manipulate money and credit through three tools — open-market operations, the policy rate, and reserve requirements

OMO is the day-to-day workhorse in developed markets. Reserve requirements are rarely changed in the US/EZ but are still actively used in China.

Tool	Mechanism	Example
Open-market operations (OMO)	Buy/sell govt securities → inject or drain reserves, moves overnight rate toward target	Fed Open Market Desk
Policy rate (repo rate)	Rate at which CB lends short-term money to banks — anchors the yield curve short end	BoE 2-week repo · ECB MRO
Reserve requirements	Minimum share of deposits banks must hold — higher ratio → less lending capacity	PBoC RRR tool

UNCONVENTIONAL · QE

Large-scale asset purchases flatten the yield curve when short rates hit zero. Balance sheets of Fed, ECB, BoJ ballooned 4–8× post-2008.

UNCONVENTIONAL · NIRP

Negative interest-rate policy: ECB (2014–22) and BoJ (2016–24) charged banks on excess reserves to push lending.

UNCONVENTIONAL · FORWARD GUIDANCE

"Rates will stay low until inflation is durably at 2%" — shapes expectations without changing the current rate.

Source: CFA Institute, Curriculum Vol 2, Learning Module 4, §1 "Monetary Policy Tools and Transmission", pp. 118–132.

The policy rate propagates through four channels — short rates, asset prices, the exchange rate, and expectations — to influence real activity and inflation

Full transmission takes 12–24 months. This lag is why central banks must act on forecasts of inflation, not realized inflation.

1 **Short-term interest rates**
Policy rate → interbank rate → bank lending rates → corporate & household borrowing costs → consumption and investment.

2 **Asset prices**
Lower rates raise equity, bond, and real-estate prices → wealth effect on consumption; lower cost of capital for firms.

3 **Exchange rate**
Lower domestic rate → weaker currency → higher net exports and imported inflation; stronger currency works in reverse.

4 **Expectations**
A credible central bank shifts the inflation expectations of firms and households, which feed into wage- and price-setting.

Source: CFA Institute, Curriculum Vol 2, Learning Module 4, §1 "Monetary Policy Tools and Transmission", pp. 128–132.

Inflation targeting anchors expectations by committing the central bank to a public, numerical inflation goal — now the global standard

The New Zealand Reserve Bank pioneered a 2% inflation target in 1990. By 2025, 40+ central banks have adopted a formal numerical target.

WHY 2% · NOT 0%

A small positive target creates a buffer against deflationary traps, allows real wages to adjust downward when needed, and avoids the zero lower bound on nominal rates.

CREDIBILITY · THE ANCHOR

Transparent commitment + operational independence + accountability framework. Once inflation expectations anchor at 2%, wage- and price-setting behave as if inflation will be 2%.

Central bank	Target	Mandate type
ECB	"Below, but close to, 2%" HICP	Price stability (primary)
Bank of England	2% CPI	Price stability (primary)
RBNZ (pioneer)	1–3% CPI, midpoint 2%	Price stability
US Federal Reserve	2% PCE (flexible, avg)	DUAL — prices + max employment
Bank of Japan	2% CPI (no hard target historically)	Price stability

The Taylor rule implies a policy rate equal to the neutral rate plus a weighted response to inflation and output gaps — the workhorse reaction function

Neutral rate ≈ real trend g + inflation target. Above this rate policy is contractionary; below it, expansionary.

$$i^* = r_neutral + \pi^* + 0.5 \times (\pi - \pi^*) + 0.5 \times (y - y^*)$$

where: i^* = implied policy rate; r_neutral = neutral real rate; π^* = inflation target; $(\pi - \pi^*)$ = inflation gap; $(y - y^*)$ = output gap (% of potential).

TAYLOR RULE — IMPLIED POLICY RATE CALCULATION

Assume real trend growth g = 2.5%, inflation target π^* = 2%, observed inflation π = 3%, output gap $y - y^*$ = +1%. What policy rate does the Taylor rule prescribe?

1. Neutral nominal rate = r_neutral + π^* = 2.5% + 2.0% = 4.5%.
2. Inflation gap response: 0.5 × (3% − 2%) = 0.5%.
3. Output gap response: 0.5 × (+1%) = 0.5%.
4. i^* = 4.5% + 0.5% + 0.5% = 5.5%.

Implied policy rate = 5.50% — tighter than neutral, because both inflation and output are above target.

[1] The original Taylor (1993) weights are 0.5 on each gap. Modified Taylor rules use higher inflation weights (1.5) to satisfy the Taylor principle.

Source: CFA Institute, Curriculum Vol 2, Learning Module 4, §2 "The Neutral Rate", pp. 138–142; Taylor (1993).

SECTION 03

Geopolitics for investors

State actors cooperate or compete along a spectrum driven by national security, economic interest, and soft-power considerations

Cooperation emerges where the national interest aligns — rules standardization, free movement of goods/people/capital, technology exchange. Its opposite: sanctions, tariffs, capital controls, retaliation.

STATE ACTORS

National governments, central banks, finance ministries, legislatures — entities that wield formal sovereignty over security and resources.

NON-STATE ACTORS

NGOs, multinationals, rating agencies, media, influential individuals — no sovereignty but material influence over flows, narratives, and standards.

Cooperation driver	Typical action	Example
National security	Alliance treaties, joint exercises	NATO Article 5
Economic interest	Free-trade agreements, tariff harmonization	EU single market · USMCA
Resource endowment	Strategic partnerships on food, energy, minerals	IEA · Saudi–US oil pact
Standardization	Common protocols reduce friction	Basel III · SWIFT · ISO
Soft power	Cultural export, education exchange, tourism	Erasmus+ · K-pop diplomacy

Source: CFA Institute, Curriculum Vol 2, Learning Module 5, §1 "Introduction to Geopolitics", pp. 144–156.

Geopolitical risk for investors splits into three types — event, exogenous, and thematic — each demanding a different monitoring cadence

The BlackRock/Caldara GPR index shows high geopolitical risk depresses investment, employment, and equity prices — the threat is often more damaging than the event itself.

EVENT RISK

Date-driven, calendar-observable.

Elections, referenda, summits, legislative deadlines, policy reviews.

Example: UK 2016 Brexit referendum — GBP dropped 8% in one session.

Monitoring: political calendar, week-level.

EXOGENOUS RISK

Sudden, unanticipated shocks.

Invasions, coups, terrorist attacks, natural disasters, pandemic onset.

Example: 2022 Russia–Ukraine invasion — European gas tripled in weeks.

Monitoring: tail-risk scenarios, hedging overlays.

THEMATIC RISK

Known, slow-moving, evolving.

Climate change, populism, migration, US–China strategic rivalry, cyber.

Example: 2018– US–China tariff cycle — 40% of trade reshored or diverted.

Monitoring: strategic review, quarter-level.

Source: CFA Institute, Curriculum Vol 2, Learning Module 5, §3 "Geopolitical Risk", pp. 180–194.

SECTION 04

International trade & BoP

Countries gain from trade by specializing according to comparative advantage — the Ricardian insight is that absolute productivity is irrelevant

Heckscher–Ohlin extension: a country exports the good that intensively uses its abundant factor (capital-abundant → capital-intensive exports).

Specialize where opportunity cost is lowest → trade gains both countries

where: Opportunity cost of good X in country A = units of good Y given up to produce one more X.

TWO-COUNTRY, TWO-GOOD RICARDIAN EXAMPLE

Home produces 10 cloth OR 5 wine per hour; Foreign produces 4 cloth OR 4 wine per hour. Home has absolute advantage in both — should they still trade?

1. Home opportunity cost of 1 cloth = 0.5 wine; Foreign opp. cost of 1 cloth = 1 wine.
2. Foreign opportunity cost of 1 wine = 1 cloth; Home opp. cost of 1 wine = 2 cloth.
3. Home has comparative advantage in cloth (0.5 < 1); Foreign has comparative advantage in wine (1 < 2).
4. With trade at e.g. 1 cloth = 0.75 wine, both are strictly better off than in autarky.

Specialization + trade ⇒ combined output rises even though Home has absolute advantage in both goods.

Source: CFA Institute, Curriculum Vol 2, Learning Module 6, §2 "Benefits and Costs of Trade", pp. 196–200.

Tariffs, quotas, VERs, and export subsidies all protect domestic producers at the cost of a larger consumer-surplus loss — the net effect is a deadweight loss

Welfare math: ΔConsumer Surplus = −(A+B+C+D). ΔProducer Surplus = +A. Government revenue = +C. Net national welfare = −(B+D) — the deadweight loss triangles.

Instrument	Who collects the rent	Net welfare vs tariff
Tariff (import tax)	Importing government	Baseline: −(B+D)
Quota	Ambiguous — importer or foreign seller	Worse if rent goes abroad: −(B+C+D)
Voluntary export restraint (VER)	Exporting country captures the rent	Strictly worse: −(B+C+D)
Export subsidy	Foreign consumers benefit; home loses	Clear welfare loss for exporter

SOUTH AFRICA PAPER — 20% TARIFF WELFARE DECOMPOSITION

Pre-tariff: world price USD 5.00, domestic supply 110k tons, demand 200k tons (imports 90k). Tariff 20% raises domestic price to USD 6.00; supply rises to 130k, demand falls to 170k.

|ΔCS| = 185k · |ΔPS| = 120k · Govt revenue = 40k → Deadweight loss = 185 − 120 − 40 = USD 25,000.

Source: CFA Institute, Curriculum Vol 2, Learning Module 6, §3 "Trade Restrictions", pp. 199–212.

Regional trade blocs range from free-trade areas to full economic unions, each removing a different layer of friction

Trade creation (low-cost imports replace high-cost domestic production) is welfare-enhancing. Trade diversion (non-member low-cost imports replaced by member higher-cost imports) can be welfare-reducing.

Bloc type	Removes	Example
Free-trade area (FTA)	Internal tariffs only	USMCA (US·MX·CA)
Customs union	Internal tariffs + common external tariff	Mercosur
Common market	Above + free movement of labor & capital	EEA (EU-27 + 3)
Economic union	Above + harmonized economic policy	EU single market
Monetary union	Above + shared currency and central bank	Eurozone

TRADE CREATION vs TRADE DIVERSION

Creation: Mexican avocados replacing higher-cost California avocados inside USMCA — welfare gain. Diversion: Mexican textiles replacing lower-cost Vietnamese textiles inside USMCA because of the external tariff on Vietnam — welfare loss if diversion dominates creation.

Source: CFA Institute, Curriculum Vol 2, Learning Module 6, §4 "Trading Blocs", pp. 206–214.

The balance of payments decomposes into a current account, a capital account, and a financial account — which must sum to zero in theory

A country with a current-account deficit must fund it with a financial-account surplus — i.e. net foreign capital inflows. The US has run this pattern continuously since 1982.

$$\text{Current Account} + \text{Capital Account} + \text{Financial Account} = 0$$

where: CA captures trade, services, income, transfers. KA captures non-produced, non-financial capital. FA captures portfolio + FDI + reserves.

CURRENT ACCOUNT

Trade balance (goods) + services + primary income (dividends, interest on foreign assets) + secondary income (transfers, remittances).

US 2024 CA ≈ −3% of GDP.

CAPITAL ACCOUNT

Debt forgiveness, inheritance/migrant transfers, sale of non-produced assets (patents, land rights).

Typically <0.1% of GDP for developed economies.

FINANCIAL ACCOUNT

FDI + portfolio flows (equity, debt) + other investment (bank loans) + reserve assets (central bank FX intervention).

US 2024 FA ≈ +3% of GDP.

[1] In practice a "net errors and omissions" item closes the identity because measurement is imperfect.

Source: CFA Institute, Curriculum Vol 2, Learning Module 7, §2 "FX and the Trade Balance", pp. 240–254.

SECTION 05

FX market structure

The FX market is the world's largest — about USD 7.5 trillion daily turnover — and it never sleeps, with most trades involving the US dollar

Spot is a minority of volume; FX swaps (spot + offsetting forward) dominate because they roll and finance positions. Most flow is buy-side (financial clients), not interbank.

Instrument	Share of daily FX turnover
Spot	33%
Outright forwards	14%
FX swaps	49%
Options & other	5%

Top-5 pair	Share
USD/EUR	23.1%
JPY/USD	17.8%
USD/GBP	9.3%
USD/AUD	5.2%
CAD/USD	4.3%

MARKET STRUCTURE · BUY SIDE vs SELL SIDE

Sell side: tier-1 money-center banks (Deutsche, Citi, UBS, HSBC, JPM). Buy side: corporates (trade-flow hedging), real-money (pension, insurance, mutual funds), leveraged accounts (hedge funds, HFT algos), retail (10%+ of some pairs), SWFs, central banks. Trading hubs: London (~40%), New York, Tokyo, Singapore.

Source: CFA Institute, Curriculum Vol 2, Learning Module 7, §2 "The Foreign Exchange Market", pp. 216–230; BIS Triennial Survey 2022.

FX quotes follow a strict price-currency / base-currency convention — "USD/EUR 1.17" means one euro buys 1.17 US dollars

Hierarchy: EUR > GBP > AUD/NZD > USD > others. Whichever currency is higher in the hierarchy becomes the base (denominator-like). Yen rates quoted to 2dp; non-yen majors to 4dp.

A / B means: number of units of A (price currency) per one unit of B (base currency)

where: Direct quote = domestic is the price currency. Indirect = domestic is the base. Professional FX avoids "direct/indirect" language.

Market name	Professional quote	Interpretation at 1.1701 / 130.98 / 0.8752
Euro	USD/EUR = 1.1701	1 EUR buys 1.1701 USD
Dollar–yen	JPY/USD = 130.98	1 USD buys 130.98 JPY (2dp only)
Sterling	USD/GBP = 1.2850	1 GBP buys 1.2850 USD
Euro–sterling	GBP/EUR = 0.8752	1 EUR buys 0.8752 GBP
Dollar–Canada	CAD/USD = 1.3020	1 USD buys 1.3020 CAD

Source: CFA Institute, Curriculum Vol 2, Learning Module 7, §2 "Exchange Rate Quotations", pp. 228–234.

Bid-offer spreads compensate dealers for inventory and adverse-selection risk — banks buy the base currency at the bid and sell it at the offer

Electronic trading has compressed top-of-book spreads for EUR/USD to a fraction of a pip (≈USD 0.00002). Emerging-market pairs remain 10–100× wider.

MXN/USD BID-OFFER MECHANICS

A NY dealer quotes 18.8580 – 18.8600 MXN/USD to a Mexico-based client. Interpret the quote and compute the inverted USD/MXN bid-offer.

1. Base currency = USD; price currency = MXN. Dealer BUYS USD at 18.8580 MXN (client sells USD).
2. Dealer SELLS USD at 18.8600 MXN (client buys USD). Spread = 20 pips = 0.0020 MXN.
3. Invert: dealer's bid for MXN = 1/18.8600 = 0.05302 USD. Dealer's offer for MXN = 1/18.8580 = 0.05303 USD.

USD/MXN = 0.05302 – 0.05303. Rule: bid always < offer, regardless of quote direction.

WHY SPREADS EXIST · THREE FRICTIONS

(1) Inventory risk — dealer holds a position while searching for the other side. (2) Adverse selection — informed flow picks off stale quotes. (3) Operational and credit costs. Liquidity is deepest during overlapping London–New York hours (approx. 08:00–11:00 NY time).

Source: CFA Institute, Curriculum Vol 2, Learning Module 7, §2 "Exchange Rate Quotations", pp. 228–234.

A country can credibly choose any two of — fixed FX, free capital mobility, and independent monetary policy — but not all three

The Mundell–Fleming trilemma. Most flexible-FX economies (US, UK, JP) choose free capital + autonomy. Hong Kong pegs + free capital, gives up monetary autonomy. China historically picks autonomy + managed FX, restricts capital.

FREE FLOAT

Rate determined by market supply/demand.

Examples: USD, EUR, JPY, GBP, AUD.

Gain: full monetary autonomy. Cost: FX volatility.

MANAGED FLOAT

Official intervention smooths moves without defending a specific rate.

Examples: CNY, INR, BRL.

Middle ground.

HARD PEG

Rate fixed by law (currency board) or dollarization.

Examples: HKD/USD, BHD/USD, dollarized Ecuador.

Gain: stability. Cost: imports the anchor's monetary policy.

Regime spectrum	Example currency pair	Trilemma corner
Free float	USD/EUR, JPY/USD	Free K + autonomy
Managed float	CNY/USD (PBoC band)	Partial trade-off
Crawling peg	EGP/USD (pre-2022)	Limited autonomy
Currency board / hard peg	HKD/USD (7.75–7.85 band)	Free K + fixed FX
Monetary union	EUR across eurozone members	Shared autonomy, fixed internal FX

SECTION 06

FX arithmetic — IRP, PPP, cross-rates

Real exchange rates deflate nominal FX by the price-level ratio — they measure relative purchasing power, not a tradeable quote

A rising real exchange rate (in domestic-per-foreign terms) means it takes more domestic real income to buy a given foreign basket — your purchasing power abroad falls.

$$R_{d/f} = S_{d/f} \times (P_f / P_d) \qquad \%\Delta R \approx \%\Delta S + \%\Delta P_f - \%\Delta P_d$$

where: $S_{d/f}$ = nominal spot (units of domestic per one foreign); P_f, P_d = foreign / domestic price levels.

UK CONSUMER BUYING EUROZONE GOODS

Over one year, nominal $S_{GBP/EUR}$ rises 10%, Eurozone CPI rises 5%, UK CPI rises 2%. How much does the real cost of EZ goods change for a UK consumer?

1. $\%\Delta R_{GBP/EUR} \approx \%\Delta S + \%\Delta P_{EUR} - \%\Delta P_{UK}$
2. = +10% + 5% − 2% = +13%
3. The UK consumer pays 13% more in real terms for the same EZ basket.
4. Equivalent statement: UK real purchasing power over EZ goods fell ≈ 13%.

ΔR ≈ +13% → UK real PP over EZ goods down ~13%.

Source: CFA Institute, Curriculum Vol 2, Learning Module 7, §2 "Real Exchange Rates", pp. 218–224.

Cross-rates between two non-USD currencies are derived by chaining through the USD quote and inverting where needed

Always check the quote convention. To get X/Y from A/Y and A/X, you multiply (A/Y) × (X/A) = X/Y — i.e. one of the two must be inverted.

$$\text{CAD/EUR} = (\text{CAD/USD}) \times (\text{USD/EUR})$$

where: Units check: (CAD / USD) × (USD / EUR) → CAD / EUR. The USDs cancel, leaving the desired cross.

USD/EUR 1.1701 and CAD/USD 1.3020 → COMPUTE CAD/EUR

Spot USD/EUR = 1.1701 (1 EUR buys 1.1701 USD). Spot CAD/USD = 1.3020 (1 USD buys 1.3020 CAD). What is the CAD/EUR cross-rate?

1. Desired = CAD per 1 EUR. Chain: EUR → USD → CAD.
2. 1 EUR = 1.1701 USD → 1.1701 USD × 1.3020 CAD/USD
3. CAD/EUR = 1.3020 × 1.1701 = 1.5236
4. Sanity check: a European tourist exchanges 1 EUR, expects ~CAD 1.52 — higher than each single leg because both currencies are weak vs EUR.

CAD/EUR = 1.5236 (1 EUR buys 1.5236 CAD)

Source: CFA Institute, Curriculum Vol 2, Learning Module 8, §1 "Cross-Rate Calculations", pp. 254–258.

When a dealer's cross-rate drifts from the chain-implied value, triangular arbitrage locks in a riskless profit

In liquid markets algo desks arb the gap in milliseconds. In illiquid EM pairs, windows can persist — hence the CFA emphasis.

CFA LM 8 PRACTICE — ZAR/SEK TRIANGULAR ARB

Dealer A quotes CNY/ZAR = 0.9149 and CNY/SEK = 1.0218, implying ZAR/SEK = 1.0218 / 0.9149 = 1.1168. Dealer B quotes ZAR/SEK = 1.1210. Construct the arbitrage on SEK 1 million.

1. Implied ZAR/SEK via Dealer A = 1.1168 (lower — SEK is cheaper in ZAR via A).
2. Dealer B quote 1.1210 is HIGHER — Dealer B pays more ZAR per SEK.
3. Buy SEK from A at 1.1168 (pay ZAR 1,116,800 for SEK 1m) → sell SEK to B at 1.1210 (receive ZAR 1,121,000).
4. Profit = SEK 1,000,000 × (1.1210 − 1.1168) = ZAR 4,200.

Riskless profit = ZAR 4,200 per SEK 1 million, no capital at risk.

ARBITRAGE DISCIPLINE · CHECK THE CHAIN BEFORE YOU TRADE

Step 1 — compute chain-implied cross from the liquid quotes. Step 2 — compare to any direct cross quote. Step 3 — if gap exceeds round-trip transaction cost, execute in the direction that buys the currency cheap in one venue and sells it dear in the other. Close the triangle.

Source: CFA Institute, Curriculum Vol 2, Learning Module 8, §1 Practice Problem, pp. 254–258.

Forward exchange rates are pinned by a no-arbitrage condition linking spot, forward, and the two risk-free rates — covered interest-rate parity

Forward points are scaled differences (F − S). Non-yen majors × 10,000 (4 dp). Yen × 100 (2 dp). Day count Actual/360 for most Libor/SOFR conventions.

$$F_{f/d} = S_{f/d} \times (1 + r_f \times \tau) / (1 + r_d \times \tau) \qquad F_{f/d} - S_{f/d} = S_{f/d} \times (r_f - r_d) \times \tau / (1 + r_d \times \tau)$$

where: Quote f/d: units of foreign per one domestic; τ = actual days / 360; r_d, r_f = domestic / foreign risk-free rates.

Condition	Base currency is trading at	Interpretation
r_f > r_d (foreign rate higher)	Forward PREMIUM (F > S)	Arbitrage equalizes returns
r_f < r_d (foreign rate lower)	Forward DISCOUNT (F < S)	Base currency forwards cheaper
r_f = r_d	F = S (flat points)	No interest-rate wedge

TESTABLE RULE

The currency with the HIGHER interest rate always trades at a forward DISCOUNT vs the lower-rate currency — otherwise a riskless arbitrage exists. This single rule answers most CFA IRP questions.

Source: CFA Institute, Curriculum Vol 2, Learning Module 8, §2 "Forward Rate Calculations", pp. 258–264.

Covered interest-rate-parity violations open a riskless arbitrage — borrow in the low-yield currency, swap, lend in the high-yield currency, unwind via the forward

Real-world CIP deviations (widen post-2008 because of bank-balance-sheet costs) are measured by the "cross-currency basis" — a Nobel-worthy empirical puzzle.

CFA LM 8 — COVERED IRP ARBITRAGE

Spot $S_{f/d}$ = 1.6535. Domestic 12m rate r_d = 3.50%. Foreign 12m rate r_f = 5.00%. A dealer misquotes the 12m forward at F = 1.6900 (vs correct F_{theor} = 1.6535 × 1.05 / 1.035 = 1.6775).

1. Check IRP: theoretical $F_{f/d}$ = 1.6535 × (1.05/1.035) = 1.6775.
2. Dealer quote 1.6900 > 1.6775 → forward is over-priced (foreign currency too strong in the forward).
3. Trade: BORROW 1 unit in foreign currency at 5%, sell it spot for 1/1.6535 = 0.6048 domestic units, invest at 3.5% → 0.6048 × 1.035 = 0.6260 in 1 year.
4. Repurchase foreign via forward at 1.6900 → 0.6260 × 1.6900 = 1.0580 foreign — repay 1.05 owed, keep 0.0080.

Riskless profit ≈ 0.77% of notional (3.50% − 2.73% swap-financed return) per unit — no capital committed.

WHY THE OPPORTUNITY CLOSES

As arbitrageurs borrow the foreign currency and sell it spot, they push the spot down and the forward up until F satisfies IRP. In real markets, this adjustment takes milliseconds and the wedge is capped by transaction costs — but the threat of arbitrage is what anchors forwards to the rate differential.

Source: CFA Institute, Curriculum Vol 2, Learning Module 8, §2 Forward Rate Arbitrage Example, pp. 260–264.

Purchasing power parity posits that identical goods should cost the same everywhere — a useful long-run anchor, but a poor short-run predictor

The Economist's Big Mac index — a single standardized good — routinely shows 20–50% deviations from PPP across major economies. Arbitrage is blocked by non-tradability.

Absolute PPP: $S_{d/f} = P_d / P_f$

Relative PPP: $\%\Delta S_{d/f} \approx \pi_d - \pi_f$

where: Absolute PPP requires identical baskets and no frictions. Relative PPP only constrains changes — cannot pin the level.

WHY PPP FAILS SHORT RUN

- Non-tradable goods (haircuts, housing, local services) dominate most CPI baskets
- Transport costs and tariffs (> 10% for many goods)
- Baskets differ across countries
- Capital flows are 50× trade flows and ignore PPP

WHEN PPP MATTERS

- Over 10+ year horizons real FX reverts (half-life ~4 years)
- In high-inflation regimes, relative PPP pins nominal FX tightly
- For back-of-envelope "fair-value" screens on EM currencies
- Big Mac / iPhone indices as communication tools

Source: CFA Institute, Curriculum Vol 2, Learning Module 7, §2 "Purchasing Power Parity", pp. 218–222.

The currency with the higher interest rate trades at a forward discount — this is the single most-tested FX relationship in CFA Level 1

Short drills for intuition — memorise "base currency with higher rate ⇒forward discount" and you already pass 80% of FX questions.

Spot	r_d (domestic)	r_f (foreign)	Forward 12m	Points
USD/EUR = 1.1000	5.00%	2.00%	1.1000 × 1.02/1.05 = 1.0686	−314
GBP/EUR = 0.8500	4.00%	3.00%	0.8500 × 1.03/1.04 = 0.8418	−82
JPY/USD = 150.00	0.25%	5.00%	150 × 1.0025/1.05 = 143.22	−678
CHF/EUR = 0.9500	1.50%	3.50%	0.9500 × 1.035/1.015 = 0.9687	+187

THREE-LINE IRP CHEATSHEET

1. Identify which is base (denominator): the "B" in A/B. 2. Compare r_base vs r_price: if r_base > r_price ⇒forward DISCOUNT (F < S); if <, forward PREMIUM. 3. Compute F = S × (1 + r_price·τ) / (1 + r_base·τ). Points = (F − S) × 10,000 (× 100 for yen).

[1] Drills use 4-decimal quoting and annual rates. Points: Δ × 10,000 for non-yen, × 100 for yen.

Source: CFA Institute, Curriculum Vol 2, Learning Module 8, §2 "Arbitrage Relationships", pp. 260–266.

KEY TAKEAWAYS

Deck 5 — Macro Policy, Geopolitics, Trade & FX

- Fiscal multiplier = 1 / [1 − c(1 − t)]; at c = 0.9 and t = 0.2 it is 3.57 in slack — collapses toward 1 at full employment.
- Debt-to-GDP is sustainable when real g > real r; otherwise the primary surplus must absorb the (r − g) drag.
- Central banks use OMO + policy rate + reserve requirements; transmission runs via short rates, asset prices, FX, expectations — 12–24 months lag.
- Taylor rule $i^* = r_{neutral} + \pi^* + 0.5(\pi - \pi^*) + 0.5(y - y^*)$ — benchmark for evaluating whether a central bank is tight or loose.
- Comparative advantage drives specialization regardless of absolute productivity; tariffs create deadweight loss = (B + D) triangles.
- Balance of payments identity: Current Account + Capital Account + Financial Account = 0 — a CA deficit must be financed by FA inflows.
- FX quoting convention A/B: units of A (price) per one unit of B (base). Cross-rates chain through USD; triangular arbitrage closes any gap.
- Covered IRP: $F/S = (1 + r_{price} \cdot \tau) / (1 + r_{base} \cdot \tau)$; currency with the higher interest rate always trades at a forward DISCOUNT.